Experiment 5

Fourier Series Analysis of Periodic Signals

OBJECTIVE

i) To analyze Fourier series of continuous-time periodic signals and analyze their waveform

characteristics.

(ii) To determine the Fourier coefficients and harmonic components of periodic signals.

(iii) To generate and analyze the amplitude and phase spectra of periodic signals.

(iv) To reconstruct periodic signals using Fourier series and verify the reconstructed waveform

with the original signal.

i) To analyze Fourier series of continuous-time periodic signals and analyze theirwaveform Characteristics.

Experimental Procedure

1. Define the time vector and fundamental frequency.

2. Generate the half-wave rectified sinusoidal periodic signal.

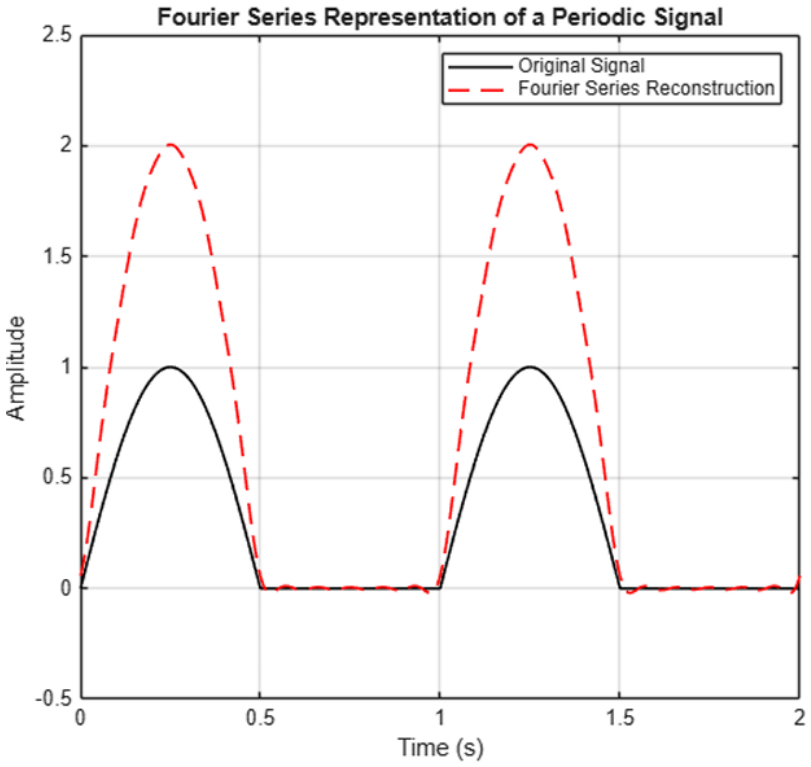
3. Compute the Fourier coefficients up to N = 10.

4. Construct the Fourier-series approximation.

5. Plot the original and reconstructed signals and compare them.

```
figure;
% Fourier Series Representation of a Periodic Signal
A = 1;
f0 = 1;
T = 1/f0;
t = linspace(0, 2*T, 2000);
% Half-wave rectified sinusoidal signal
x = A*sin(2*pi*f0*t);
x(x < 0) = 0;
N = 10;
a0 = (2/T)*trapz(t, x)/2;
an = zeros(1,N);
bn = zeros(1,N);
for k = 1:N
    an(k) = (2/T)*trapz(t, x.*cos(2*pi*k*f0*t));
    bn(k) = (2/T)*trapz(t, x.*sin(2*pi*k*f0*t));
end

xFS = a0*ones(size(t));
for k = 1:N
    xFS = xFS + an(k)*cos(2*pi*k*f0*t) ...
        + bn(k)*sin(2*pi*k*f0*t);
end
plot(t,x,'k','LineWidth',1.2); hold on;
plot(t,xFS,'r--','LineWidth',1.2);
grid on;
xlabel('Time (s)'); ylabel('Amplitude');
title('Fourier Series Representation of a Periodic Signal');
legend('Original Signal','Fourier Series Reconstruction');
```



(iv) To reconstruct periodic signals using Fourier series and verify the reconstructed waveform with the original signal.

Experimental Procedure

1. Generate the periodic signal over two periods.
2. Compute Fourier coefficients from one period.
3. Construct the reconstructed signal using 10 harmonics.
4. Calculate the error signal.
5. Plot the original, reconstructed and error signals.

Editor

ss.m | exp_3_1_ss.m | exp_4_1_ss.m | exp_5_1_ss.m | exp_5_4_ss.m *

/MATLAB Drive/exp_5_4_ss.m

```
clc; clear; close all;
%% 5(iv) Verification of Fourier Series Reconstruction and Analysis of Signal
Characteristics
figure;
% Fourier Series Reconstruction and Error Analysis
f0 = 1;
T = 1/f0;
t = linspace(0,2*T,4000);
x = max(sin(2*pi*f0*t),0);
N = 10;
a0 = (2/T)*trapz(t(1:2000),x(1:2000));
an = zeros(1,N); bn = zeros(1,N);
tt = t(1:2000); xx = x(1:2000);
for k = 1:N
    an(k) = (2/T)*trapz(tt,xx.*cos(2*pi*k*f0*tt));
    bn(k) = (2/T)*trapz(tt,xx.*sin(2*pi*k*f0*tt));

end
xr = a0/2*ones(size(t));
for k = 1:N
    xr = xr + an(k)*cos(2*pi*k*f0*t) + bn(k)*sin(2*pi*k*f0*t);
end
err = x-xr;
subplot(2,1,1);
```

